



KWAZULU-NATAL PROVINCE

EDUCATION
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 10

MATHEMATICS PAPER 2
NOVEMBER 2025

MARKS : 100

TIME: 2 hours

This question paper consists of 11 pages, including an information sheet.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of 9 questions.
2. Answer ALL the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. which you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers correct to TWO decimal places, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1

The table below shows the number of public schools from the NINE provinces in South Africa.



Provinces	Number of public schools
Eastern Cape	5 468
Free State	1 214
Gauteng	x
KwaZulu-Natal	5 895
Limpopo	3 867
Mpumalanga	1 725
Northern Cape	544
North West	1 472
Western Cape	1 450
Total	23 718

- 1.1 Determine the value of x . (1)
- 1.2 Calculate the mean number of schools in South Africa. (2)
- 1.3 Write down the median province. (1)
- 1.4 Determine the interquartile range. (3)
- 1.5 Due to the population increase in South Africa, 0,68% more schools will be required in the next 15 years. Determine the number of schools needed in the next 15 years. (2)

[9]**QUESTION 2**

The following table represents the grade 10 mathematics test results at Excel Secondary School.

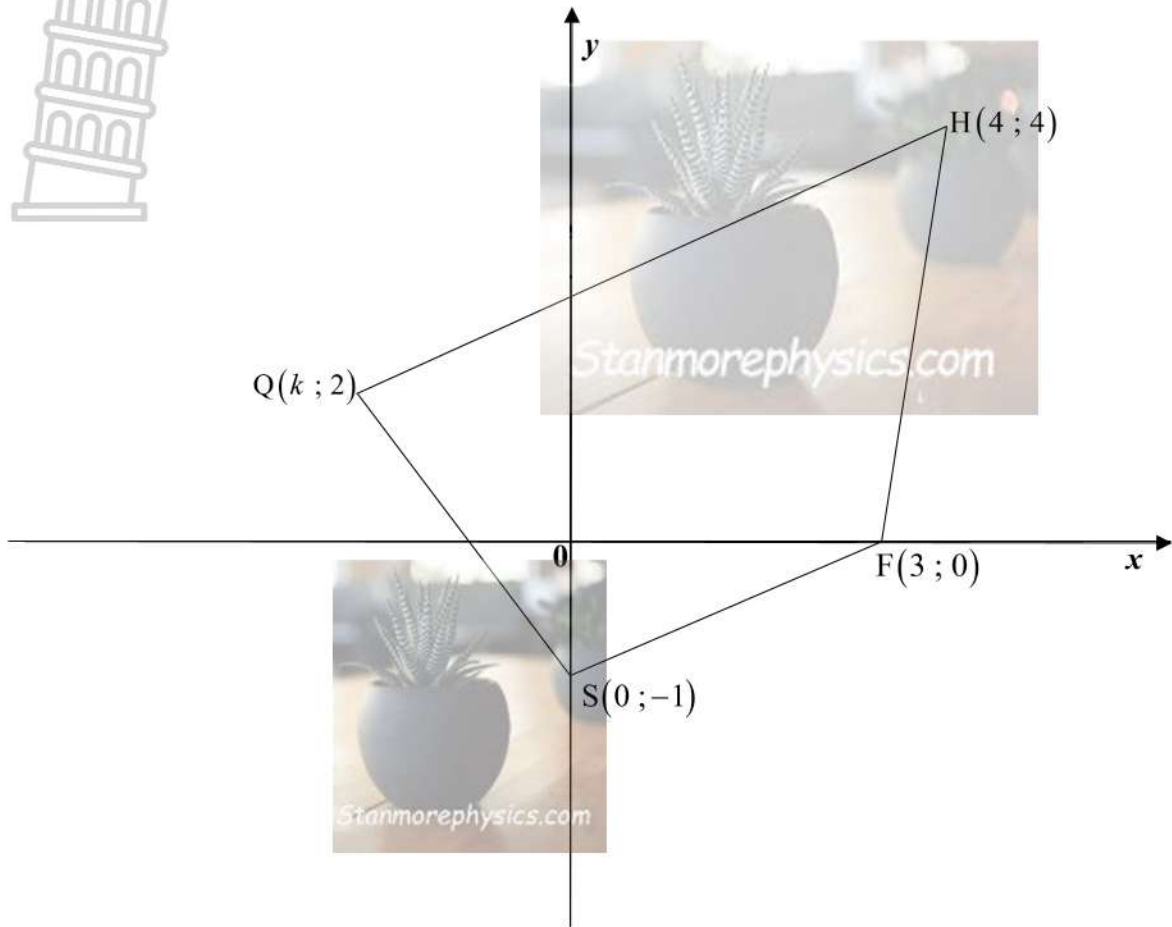
MARKS (in %)	FREQUENCY
$20 \leq x < 30$	6
$30 \leq x < 40$	11
$40 \leq x < 50$	9
$50 \leq x < 60$	7
$60 \leq x < 70$	8
$70 \leq x < 80$	13
$80 \leq x < 90$	6

- 2.1 How many learners wrote the test? (1)
- 2.2 Calculate the estimated mean. (4)
- 2.3 Calculate the percentage of learners who obtained less than SIXTY PERCENT. (2)

[7]

QUESTION 3

3.1 In the diagram below, trapezium QHFS is drawn, such that $QH \parallel SF$.



3.1.1 Determine the gradient of SF. (2)

3.1.2 Determine the equation of QH. (3)

3.1.3 Calculate the value of k . (2)

3.1.4 Determine the length of HF. (2)

3.1.5 Calculate the coordinates of M, the midpoint of HS. (3)

3.2 Line segment AB is perpendicular to the segment CD. A(-5; 2) and B(p; -1) are points on AB. If the gradient of segment CD is 2, calculate the value of p . (3)

[15]

QUESTION 4

4.1 If $x = 67,7^\circ$ and $y = 52,3^\circ$. Calculate the following correct to TWO decimal places.

4.1.1 $\frac{\sin x}{2}$ (2)

4.1.2 $\cos(3y)$ (2)

4.1.3 $\operatorname{cosec}\left(\frac{x-y}{2}\right)$ (2)

4.2 Solve for θ in the following equations (correct to ONE decimal place).

4.2.1 $3 + \cos 4\theta = 3,706$ (3)

4.2.2 $4 \tan \theta = 7 - 4 \tan \theta$ (3)

4.3 In which quadrant does the terminal arm of the angle θ lie if:

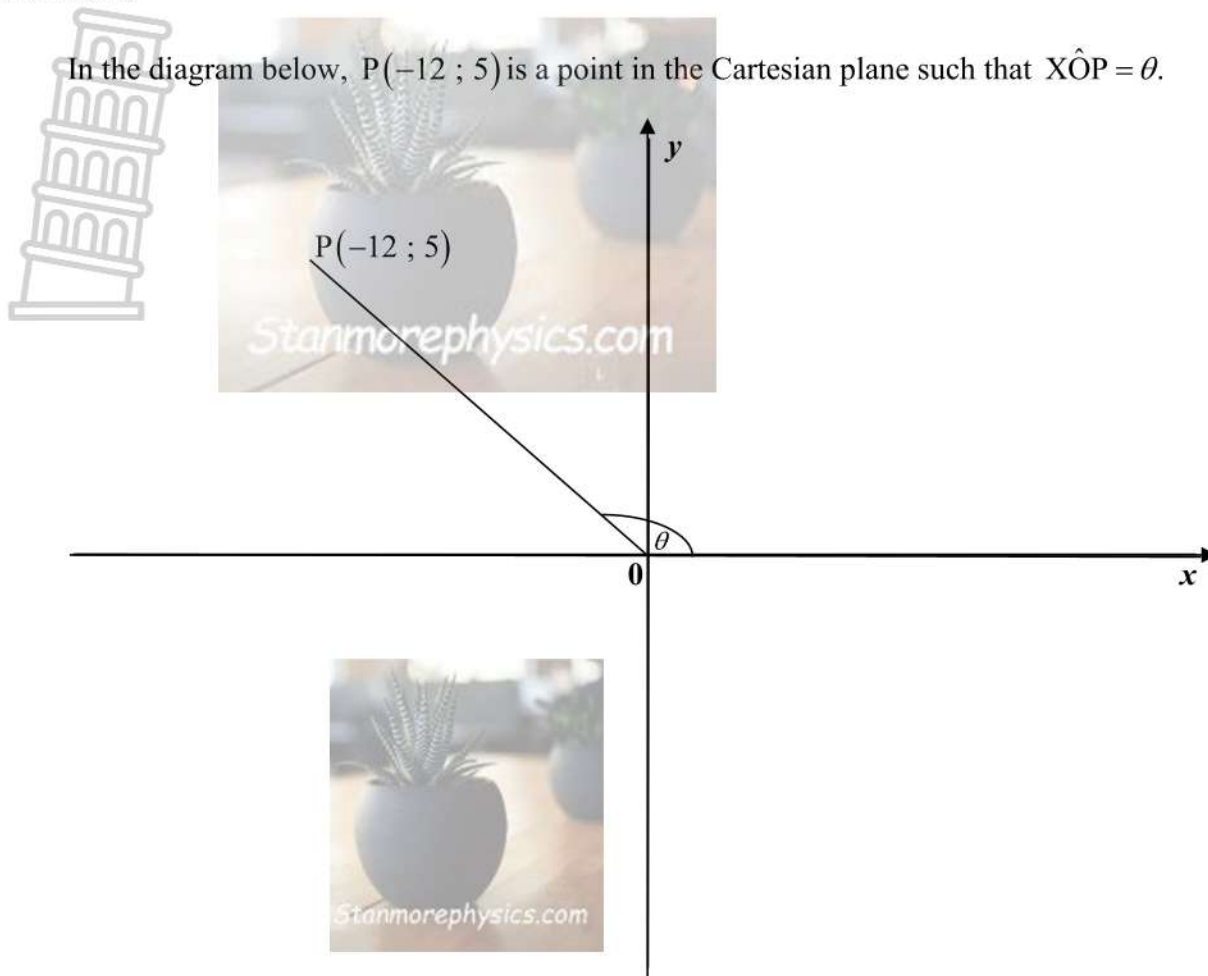
4.3.1 $\sin \theta < 0$ and $\cos \theta < 0$ (1)

4.3.2 $\tan \theta < 0$ and $\cos \theta > 0$ (1)

[14]

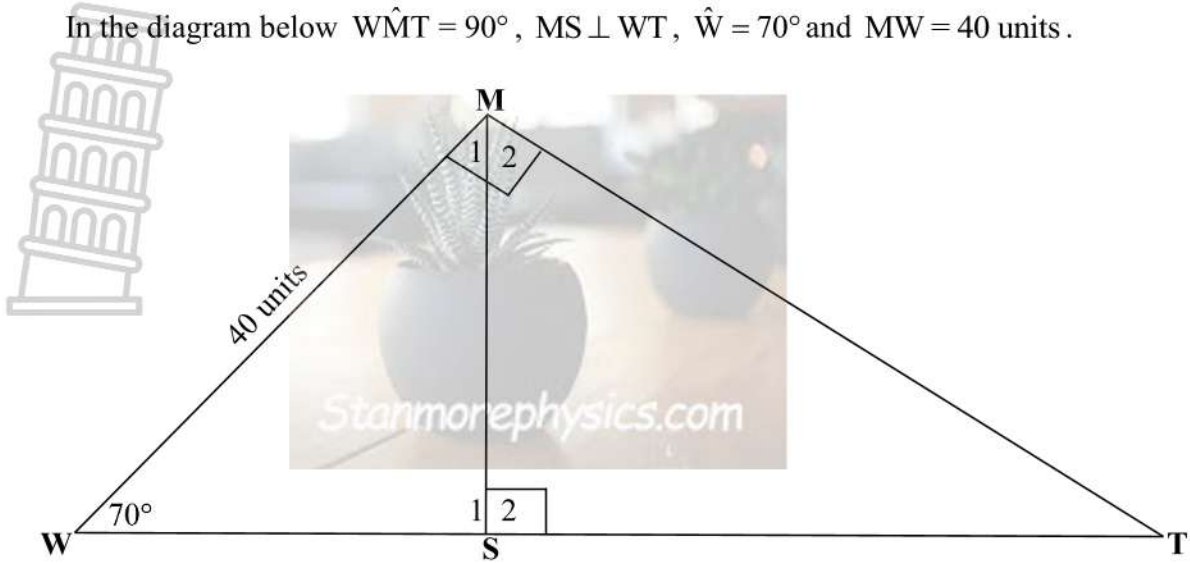
QUESTION 5

- 5.1 In the diagram below, $P(-12 ; 5)$ is a point in the Cartesian plane such that $\hat{XOP} = \theta$.



- 5.1.1 Calculate the length of OP. (2)
- 5.1.2 Hence, WITHOUT the use of a calculator, determine the value of:
- a) $\tan \theta$ (1)
- b) $(\sin \theta + \cos \theta)^2$ (3)

- 5.2 In the diagram below $\widehat{WMT} = 90^\circ$, $MS \perp WT$, $\widehat{W} = 70^\circ$ and $MW = 40$ units.



- 5.2.1 Determine the length of WS . (3)

- 5.2.2 Hence, or otherwise determine the length of ST . (5)

[14]

QUESTION 6

Given: $g(x) = \cos x + 1$ and $f(x) = 2 \sin x$ for $x \in [0^\circ; 360^\circ]$.

- 6.1 On the grid provided in your SPECIAL ANSWER BOOK, sketch the graphs of f and g on the same set of axes for $x \in [0^\circ; 360^\circ]$. (4)

- 6.2 Write down the period of f . (1)

- 6.3 Determine the range of g . (2)

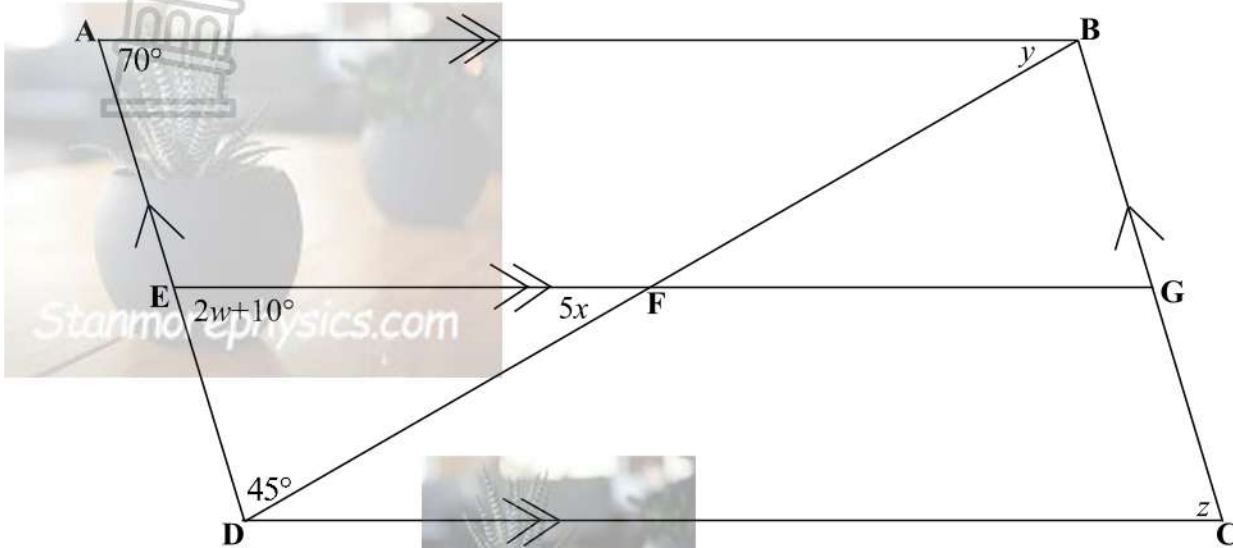
- 6.4 For which value of x will $\frac{g(x) - f(x)}{3} = 1$? (2)

[9]

In the next TWO questions, ensure you give reasons for each statement you make.

QUESTION 7

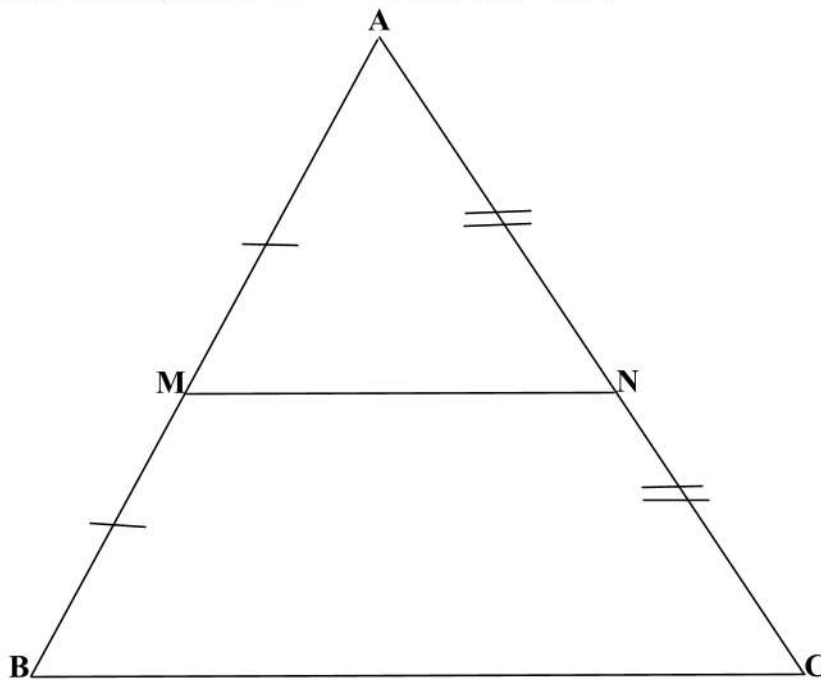
7.1 In the diagram below, ABCD is a parallelogram with $AB \parallel EG \parallel DC$.



7.1.1 Determine the values of w and x . (4)

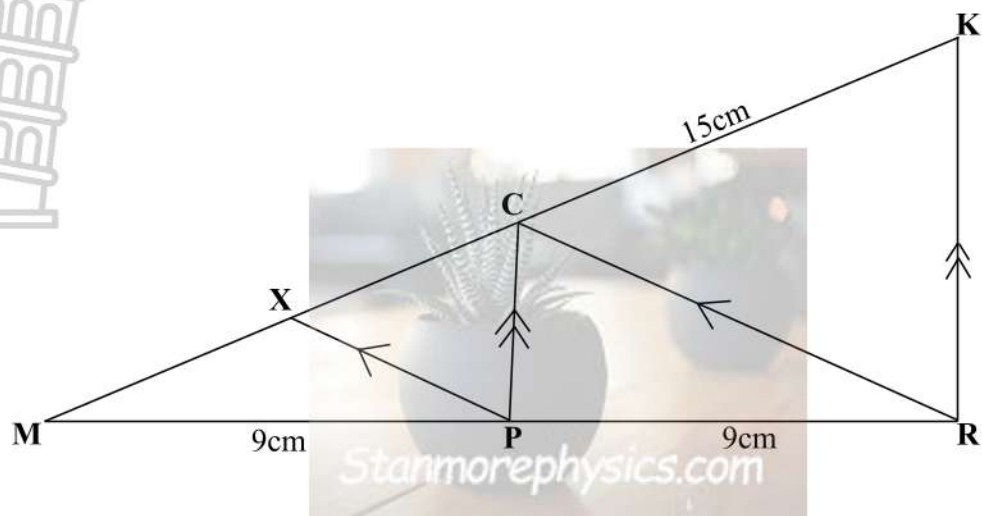
7.1.2 Calculate the values of y and z . (4)

7.2 Given $\triangle ABC$, where $AM = MB$ and $AN = NC$.



Prove the theorem which states that $MN = \frac{1}{2} BC$ (5)

- 7.3 In the diagram below $KR \parallel CP$, $PX \parallel RC$, $MP = PR = 9\text{cm}$ and $CK = 15\text{cm}$.



7.3.1 Prove that $MX = \frac{15}{2}\text{cm}$ (3)

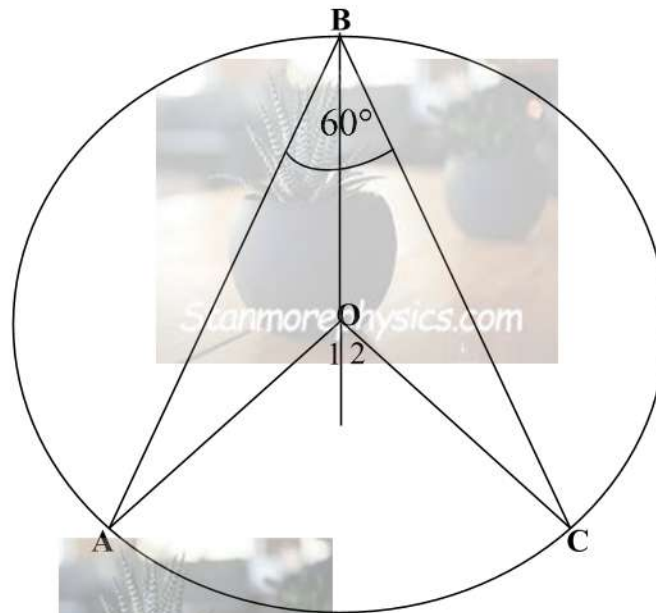
7.3.2 Calculate KR if $KR = \frac{16}{5}MX$ (2)

7.3.3 Hence, prove that $\hat{MRK} = 90^\circ$ (2)

[20]

QUESTION 8

In the diagram below, OB bisects $\hat{ABC}$. O is the centre of the circle and $\hat{ABC} = 60^\circ$.



- 8.1 Determine the size of $\hat{ABO}$. (1)
- 8.2 Calculate the size of $\hat{AOB}$. (3)
- 8.3 Hence, prove that $\triangle OAB \equiv \triangle OCB$. (3)

[7]

QUESTION 9

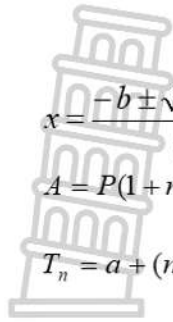
A globe of the world, spherical in shape, is placed in a geography classroom. The globe has a radius of 21 cm.

- 9.1 Calculate the volume of the globe, correct to ONE decimal place. (2)
- 9.2 Determine the area of the globe, correct to ONE decimal place. (2)
- 9.3 If the radius is doubled, what would be the ratio of the volume of the original sphere to the volume of the new sphere? (1)



[5]

TOTAL MARKS: 100

INFORMATION SHEET: MATHEMATICS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

In $\triangle ABC$:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2} ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$



KWAZULU-NATAL PROVINCE
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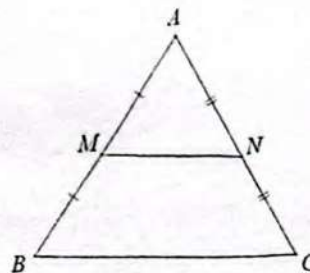
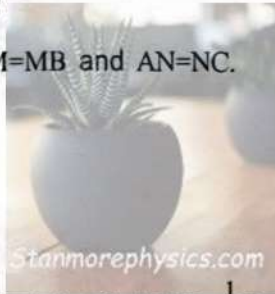
TO: DIRECTOR: CURRICULUM GRADES 10 – 12
DISTRICT DIRECTORS
DISTRICT HEADS OF CURRICULUM GRADES: 10 – 12
DISTRICT HEADS OF EXAMINATIONS AND ASSESSMENT
DISTRICT HEADS OF CIRCUIT MANAGEMENT
CIRCUIT MANAGEMENT
MATHEMATICS SUBJECT ADVISORS
PRINCIPALS OF SECONDARY SCHOOLS

**ADDENDUM TO THE PROVINCIAL COMMON EXAMINATION FOR GRADE 10
MATHEMATICS PAPER 2 NOVEMBER 2025**

Please note that question 7.2 from this question paper has to be removed and replaced as shown below:

Remove question 7.2:

Given $\triangle ABC$, where $AM=MB$ and $AN=NC$.

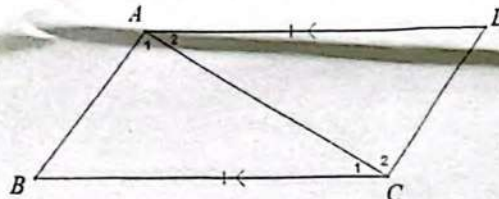


(5)

Prove the theorem which states that $MN = \frac{1}{2} BC$

Replace question 7.2 with:

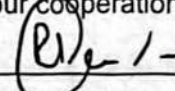
Given quadrilateral ABCD as shown below, with $AD=BC$ and $AD \parallel BC$. Prove that ABCD is a parallelogram.



(5)

PLEASE NOTE: THIS QUESTION MUST BE ANSWERED ON PAGE 16 UNDER ADDITIONAL SPACE

Your cooperation in this regard is highly appreciated.


DR. R.C. PENNISTON
CHIEF DIRECTOR EXAMINATIONS, ASSESSMENT
AND QUALITY ASSURANCE

04.11.2025

DATE



KWAZULU-NATAL PROVINCE

EDUCATION
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 10

MATHEMATICS P2

NOVEMBER 2025

MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 10 pages.

QUESTION 1			
1.1	$x = 2083$	A✓ answer	(1)
1.2	$\text{mean} = \frac{23718}{9}$ $\text{mean} = 2635,33$ ANSWER ONLY: FULL MARKS PENALISE HERE FOR ROUNDING	A✓✓ answer	(2)
1.3	Mpumalanga ALSO ACCEPT 1725	CA✓ answer	(1)
1.4	$\text{IQR} = Q_3 - Q_1$ $\text{IQR} = 4667,5 - 1332$ $= 3335,50$	CA✓ Q_3 and Q_1 CA✓ correct subst CA✓ answer	(3)
1.5	Number of schools = $23718(0,68\%)$ $= 161,28$ ≈ 162 schools ALSO ACCEPT Number of schools = $23718(100,68\%)$ $= 23879,28$ ≈ 23880 schools	A✓ A✓ A✓ A✓	(2)
			[9]

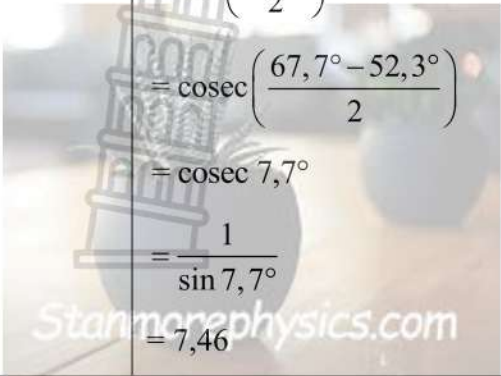
QUESTION 2			
2.1	60 learners	A ✓ answer	(1)
2.2	$\bar{x} = \frac{(25.6) + (35.11) + (45.9) + (55.7) + (65.8) + (75.13) + (85.6)}{60}$ $\bar{x} = \frac{3330}{60}$ $\bar{x} = 55,50$ ANSWER ONLY: FULL MARKS OR $\bar{x} = \frac{3330}{60}$ $\bar{x} = 55,50$	A ✓ correct midpts CA✓ 60 CA✓ 3330 CA✓ answer A ✓✓ 3330 CA✓ 60 CA✓ answer	(4)

GRADE10
Marking Guideline

2.3	$\frac{33}{60} \times 100$ $= 55\%$	A ✓ 33 CA ✓ answer	(2)
			[7]
QUESTION 3			
3.1.1	$m_{SF} = \frac{y_2 - y_1}{x_2 - x_1}$ $m_{SF} = \frac{0+1}{3-0}$ $m_{SF} = \frac{1}{3}$ ANSWER ONLY: FULL MARKS	A ✓ subst into correct formula CA ✓ answer	(2)
3.1.2	$m_{QH} = m_{SF} = \frac{1}{3}$ $y = mx + c$ $4 = \frac{1}{3}(4) + c$ $c = \frac{8}{3}$ $y = \frac{1}{3}x + \frac{8}{3}$ OR $m_{QH} = m_{SF} = \frac{1}{3}$ $y - y_1 = m(x - x_1)$ $y - 4 = \frac{1}{3}(x - 4)$ $y = \frac{1}{3}x + \frac{8}{3}$	CA ✓ $m_{QH} = m_{SF}$ CA ✓ correct subst CA ✓ answer CA ✓ $m_{QH} = m_{SF}$ CA ✓ correct subst CA ✓ answer	(3)
3.1.3	$2 = \frac{1}{3}k + \frac{8}{3}$ $\frac{1}{3}k = -\frac{2}{3}$ $k = -2$ OR	CA ✓ correct subst CA ✓ answer $k < 0$	

$$\frac{4-2}{4-k} = \frac{1}{3}$$
$$\frac{2}{4-k} = \frac{1}{3}$$
$$k = -2$$

$$\text{HF} = \sqrt{x}$$


4.1.3	 $\operatorname{cosec}\left(\frac{x-y}{2}\right)$ $= \operatorname{cosec}\left(\frac{67,7^\circ - 52,3^\circ}{2}\right)$ $= \operatorname{cosec} 7,7^\circ$ $= \frac{1}{\sin 7,7^\circ}$ $= 7,46$	A✓ correct subst	
		CA✓ answer	(2)
4.2.1	$3 + \cos 4\theta = 3,706$ $\cos 4\theta = 0,706$ $4\theta = 45,08961075^\circ$ $\theta = 11,3^\circ$ 	A✓ $\cos 4\theta = 0,706$ CA✓ $4\theta = 45,08961075^\circ$ CA✓ answer	(3)
4.2.2	$4 \tan \theta = 7 - 4 \tan \theta$ $8 \tan \theta = 7$ $\tan \theta = \frac{7}{8}$ $\theta = 41,2^\circ$ 	A✓ $8 \tan \theta = 7$ CA✓ $\tan \theta = \frac{7}{8}$ CA✓ answer	(3)
4.3.1	Third quadrant	A✓	(1)
4.3.2	Fourth quadrant	A✓	(1)
			[14]

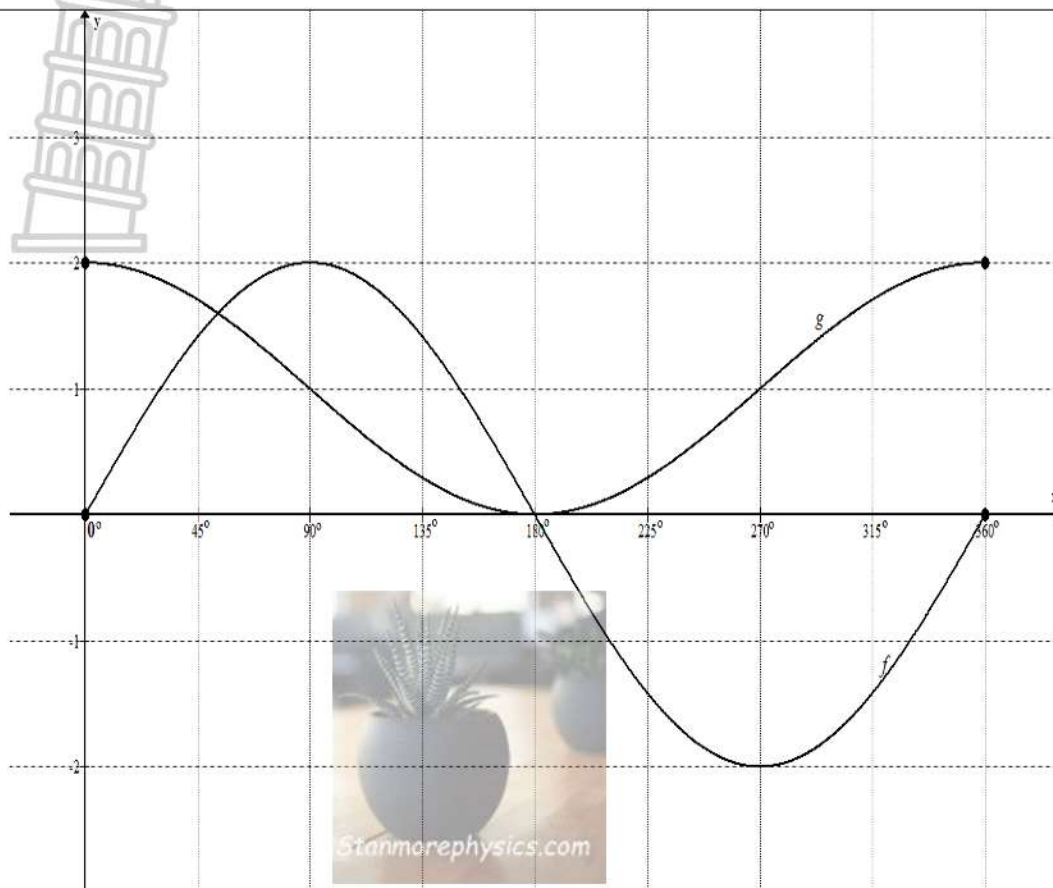
QUESTION 5

5.1.1	$x^2 + y^2 = r^2$ Pythagoras $(-12)^2 + (5)^2 = OP^2$ $OP = 13 \text{ units}$	A✓ correct subst	
	ANSWER ONLY: FULL MARKS	CA✓ answer	(2)
5.1.2	a) $\tan \theta = -\frac{5}{12}$	A✓	(1)

	b) $(\sin \theta + \cos \theta)^2$ $= \left(\frac{5}{13} + \left(\frac{-12}{13} \right) \right)^2$ $= \left(-\frac{7}{13} \right)^2$ $= \frac{49}{169}$ OR $= \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta$ $= \left(\frac{5}{13} \right)^2 + 2 \left(\frac{5}{13} \right) \left(\frac{-12}{13} \right) + \left(\frac{-12}{13} \right)^2$ $= \frac{25}{169} + \frac{-120}{169} + \frac{144}{169}$ $= \frac{49}{169}$	CA✓✓ correct subst CA✓ answer A✓ simplification CA✓ correct subst CA✓ answer	(3)
5.2.1	$\hat{S}_1 = 90^\circ$ angles on str line $\cos 70^\circ = \frac{WS}{40}$ WS = 13,68 units	A✓ $\hat{S}_1 = 90^\circ$ A✓ correct subst into trig ratio CA✓ answer	(3)
5.2.2	$\cos \hat{W} = \frac{WM}{WT}$ $\cos 70^\circ = \frac{40}{WT}$ WT = 116,952176 ST = 116,952176 – 13,68 ST = 103,27 units OR $\hat{T} = 20^\circ$ sum of Δ $\sin 20^\circ = \frac{40}{WT}$ WT = 116,952176 ST = 116,952176 – 13,68 ST = 103,27 units	A✓ trig ratio A✓ correct subst CA✓ WT = 116,952176 CA✓ difference CA✓ answer A✓ S/R CA✓ correct subst into trig ratio CA✓ WT = 116,952176 CA✓ difference CA✓ answer	
			[14]

QUESTION 6

6.1



$g(x)$

A✓ shape

A✓ Tpt

$f(x)$

A✓ shape

A✓ x int

(4)

6.2 360°

A✓

(1)

6.3 $0 \leq y \leq 2$ or $y \in [0; 2]$

CA✓
 $y \geq 0$

CA✓
 $y \leq 2$

(2)

6.4 $g(x) - f(x) = 3$
 $x = 270^\circ$

ANSWER ONLY: FULL MARKS

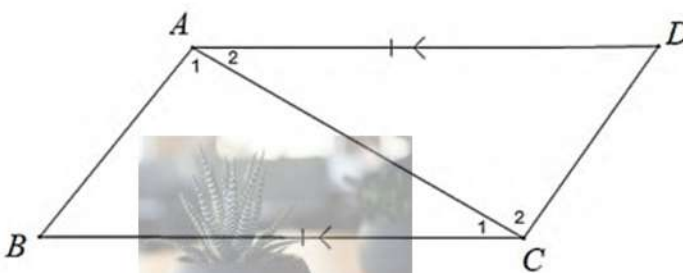
A✓

CA✓

(2)

[9]

QUESTION 7

7.1.1	$2w + 10 = 70^\circ$ $w = 30^\circ$ $5x + 45^\circ + 70^\circ = 180^\circ$ sum $\angle$ s in Δ $x = 13^\circ$	corr $\angle$ s AB//EG A✓ S/R A✓ answer A✓ S/R A✓ answer	(4)
7.1.2	$y + 70^\circ + 45^\circ = 180^\circ$ sum $\angle$ s in Δ $y = 65^\circ$ $z = 70^\circ$ opp $\angle$ s of parm	A✓ S/R A✓ answer ✓S✓ R	(4)
7.2	 In ΔABC and ΔADC AC is common $\hat{C}_1 = \hat{A}_2$ alt $\angle$s AD//BC BC = AD given $\therefore \Delta ABC \equiv \Delta ADC$ SAS $\hat{A}_1 = \hat{C}_2$ congruent Δs $\therefore AB \parallel DC$ alt $\angle$s equal $\therefore ABCD$ is a parm 2 pairs of opposite sides parallel ALTERNATIVELY AD = BC given AD // BC given $\therefore ABCD$ is a parm One pair of opposite sides equal and parallel		

(5)

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Marking Guideline

7.3.1	$MC = KC$ line through midpt // to 2nd side $MC = 15$ $MX = CX$ Line through midpt // 2nd side $MC = MX + CX$ $2MX = 15$ $MX = \frac{15}{2}$	$A\checkmark$ S/R $A\checkmark$ 15 $A\checkmark$ S/R	(3)
7.3.2	$KR = \frac{16}{5}$ $KR = \frac{16}{5} \times \frac{15}{2}$ $KR = 24$	$A\checkmark$ correct subst $A\checkmark$ answer	(2)
7.3.3	$MK^2 = 30^2$ $MK^2 = 900$ $MR^2 + KR^2 = 18^2 + 24^2$ $MR^2 + KR^2 = 900$ $\hat{MRK} = 90^\circ$ converse Theorem of Pythagoras	$A\checkmark$ $CA\checkmark$	(2)
			[20]

QUESTION 8

8.1	$\hat{ABO} = 30^\circ$	OB bisects $\hat{ABC}$	$A\checkmark$ S	(1)
8.2	$\hat{A} = \hat{ABO}$ $\hat{A} = 30^\circ$ $\hat{O}_1 = \hat{ABO} + \hat{A}$ $\hat{O}_1 = 60^\circ$ OR $\hat{A} = \hat{ABO}$ $\hat{AOB} = 120^\circ$ $\therefore \hat{O}_1 = 60^\circ$	$\angle$ s opp equal sides Ext $\angle$ s of Δ $\angle$ s opp equal sides sum of Δ $\angle$ s on a str line	$A\checkmark$ S/R $A\checkmark$ $\hat{A} = 30^\circ$ $A\checkmark$ answer $A\checkmark$ S/R $A\checkmark$ S/R $A\checkmark$ S/R	(3)

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Marking Guideline

8.3	In $\triangle ABO$ and $\triangle CBO$ 1) BO is common 2) $\hat{A}BO = \hat{C}BO = 30^\circ$ OB bisects $\hat{A}BC$ 3) $\hat{A} = \hat{A}BO = 30^\circ$ $\angle$s opp = sides $\hat{C} = \hat{C}BO = 30^\circ$ $\angle$s opp = sides $\therefore \hat{A} = \hat{C}$ $\therefore \hat{A}OB = \hat{C}OB$ sum of $\triangle$ $\triangle ABO \equiv \triangle CBO$ SAA	$A\checkmark$ S/R $A\checkmark$ S/R $A\checkmark$ SAA	(3)
			[7]

QUESTION 9

9.1	$V = \frac{4}{3}\pi r^3$ $V = \frac{4}{3}\pi(21)^3$ $V = 38792,4 \text{ cm}^3$ 	$A\checkmark$ correct subst $CA\checkmark$ answer Provided π and r used in the formula	(2)
9.2	$A = 4\pi r^2$ $A = 4\pi(21)^2$ $A = 5541,8 \text{ cm}^2$	$A\checkmark$ correct subst $CA\checkmark$ answer Provided π and r used in the formula	(2)
9.3	1:8 or $\frac{1}{8}$	$CA\checkmark$ answer	(1)
			[5]

TOTAL MARKS: 100